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The picokelvin temperature range will be approached by evaporative cooling followed by adiabatic expansion which lowers the energy even further. Although the theoretical limits of evaporative cooling are in the picokelvin range, the speed of cooling becomes extremely slow because the cooling rate is proportional to the rate of elastic collisions and therefore to the thermal velocity. However, once a Bose condensate has been created at the current critical temperatures of 100 nK to 1 microK, one might adiabatically expand the condensate and thus obtain very dilute condensates with much reduced zero-point energy and interaction energy. The major challenge is that the field curvatures have to be reduced to 10^-5 G/cm^2 while a magnetic field gradient of 8 G/cm is required to compensate for gravity in the laboratory. Such expanded Bose condensates would be coherent matter waves with de Broglie wavelengths approaching 1 mm which are not only very promising for precision metrology, but also for atom optics and conceptional experiments in quantum mechanics.

Task Description:	A systematic study of the limiting processes in generating such pK samples of atoms will be performed. Possible limitations are the precise control of the magnetic fields and heating due to background gas collisions and vibrations. We expect that these processes will become more critical at lower temperatures. Precise RF (or Raman) spectroscopy can be used to measure fundamental constants like h/m and to test fundamental symmetries like charge neutrality and time reversal. RF spectroscopy of Bose condensates will be performed, both in free fall and confined in a magnetic trap. The first steps will focus on the study of systematic shifts due to the trapping potential and the presence of the Bose condensate at finite density. results from this study will naturally lead to the formulation of a concept how to implement RF spectroscopy with mHz linewidth in the microgravity environment of a space station. The goal is to produce atomic samples with picokelvin temperatures and to improve precision spectroscopy. In microgravity, picokelvin samples would spread out by less than 1 cm/minute and would allow rf and microwave spectroscopy with mHz linewidths, an improvement of three orders of magnitude over earth-based atomic fountain experiments. RF spectroscopy of such ultracold samples will be performed to study shifts and broadening due to the finite atomic density and the presence of the trapping potential. This study should lead to a clear concept of how to perform precision spectroscopy with mHz linewidth on a space station. This holds the promise of greatly improving current atomic clocks and experiments testing fundamental symmetries in physics. This task develops key technologies for preparing picokelvin samples of atoms for precision spectroscopy. At this point, the major limitation for spectroscopy will be gravity. Picokelvin atoms in a microgravity environment holds the promise of greatly improving current atomic clocks and experiments testing fundamental symmetries in physics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	1. Quantum reflection of ultracold atoms from a solid surface Quantum reflection is a process in which a particle reflects from a potential without reaching a classical turning point. Quantum reflection requires low incident velocity or weak attraction to the surface, conditions previously realized only using liquid helium surfaces or solid surfaces at grazing incidence [1]. In this work, we demonstrated quantum reflection of ultracold sodium atoms from a solid silicon surface at normal incidence [2]. Using Bose-Einstein condensates in a weak gravitomagnetic trap to reduce the atomic motion, atoms were incident on the surface at velocities as low as 1 mm/s, corresponding to collision energies of k_B×1.5 nanokelvin. Reflectivities of 20% are in qualitative agreement with theoretical predictions. When atoms were confined in one dimension by a silicon surface, lifetime measurements indicate reflection probabilities in excess of 50%. If higher reflectivities can be obtained with low-density of thinned surfaces, new atom-optical elements based on normal reflection will become possible. 2. Observation of Feshbach resonances between two different atomic species Feshbach resonances have made it possible to control interactions in ultracold atomic gases. By tuning the magnetic field near a value where the energy of two free atoms coincides with a molecular bound state, the sign and strength of the atomic interactions can be varied. So far, all experiments on Feshbach resonances studied collisions between two atoms of the same species. We have now observed three Feshbach resonances in collisions between 6Li and 23Na atoms [3]. The resonances were identified as narrow loss features when the magnetic field was varied. The molecular states causing these resonances have been identified, and additional 6Li-23Na resonances are predicted. These resonances will allow the study of degenerate Bose-Fermi mixtures with adjustable interactions, and could be used to generate ultracold heteronuclear molecules. Ultracold polar molecules could be used for quantum computation, for studies of correlated many-body systems, and for searches for an electronic dipole moment. 3. Dynamical Instability of a Doubly Quantized Vortex in a Bose-Einstein condensate The study of topological excitations and their stability is an active frontier in the field of quantum degenerate gases. Most studies focused on vortices with one quantum of circulation. In earlier work, we created doubly quantized vortices in a Bose-Einstein condensate [4], but due to technical limitation, could not observe the predicted decay . Now we have studied the time evolution of a doubly quantized vortex state [5] and directly confirm its dynamical instability by observing that a doubly-quantized vortex core splits into two singly-quantized vortex cores [6]. The characteristic time scale of the splitting process was determined as a function of atom density and was longer at higher atomic density. The vortices were topologically imprinted into the condensate by reversing the magnetic field which created topological phases. 4. Feshbach Resonances in Fermionic 6Li We performed a detailed study of Feshbach resonances in 6Li with the goal of accurately characterizing the interaction potential of two 6Li atoms. Three new resonances in the |1> and |1> states which are p-wave resonances were observed. [7] The positions of these Feshbach resonances together with the location of a narrow s-wave resonance in the |1> + |2> mixture near 543 G were used for a precise determination of the singlet s-wave scattering length. These results, however, did not constrain the position of the broad resonance near 830 G, which also depends on the triplet scattering length. An improved measurement of its location was presented and the magnitude and the origin of possible systematic errors were discussed. This broad resonance is currently used in experiments on strongly interacting fermions. 5. Formation Time of a Fermion Pair Condensate Atomic Fermi gases close to a Feshbach resonance can explore the entire regime from weak coupling to strong interaction between the particles. When cooled below a critical temperature, fermions form bosonic pairs which can condense into the ground state of the system. The nature of the pairs depends on the interaction strength: The particles can be either tightly bound into a small molecule or they can form long-range pairs, whose size can become comparable to or even larger than the interparticle spacing, analogous to Cooper pairs of electrons. The control over the interactions between particles also offers a new way to study the dynamics of a strongly interacting many-body system: One may vary the interaction strength in a time-dependent fashion and observe how fast the system responds to that perturbation. Using this method, we studied the formation time of a condensate of fermionic atom pairs close to a Feshbach resonance [8]. The observable was the fraction of condensed molecules in the cloud after a rapid magnetic field ramp across the Feshbach resonance. The measured response time of the system was slow compared to the rapid ramp, which provides final proof that the molecular condensates observed earlier [8, 9] reflect the presence of fermion pair condensates before the ramp. 6. Coherent Molecular Optics using Sodium Dimers Coherent molecular optics was performed using two-photon Bragg scattering. [10] Molecules were produced by sweeping an atomic Bose-Einstein condensate through a Feshbach resonance [11]. Using optical standing waves of suitably chosen frequencies, sodium dimers were coherently manipulated with negligible heating or other incoherent processes. The spectral width of the molecular Bragg resonance

Task Progress:	which is Doppler sensitive corresponded to an instantaneous temperature of 20 nK, indicating that atomic coherence was transferred directly to the molecules. An autocorrelating interference technique was used to observe the quadratic spatial dependence of the phase of an expanding molecular cloud. Finally, atoms initially prepared in two momentum states were observed to crosspair with one another, forming molecules in a third momentum state. This process is analogous to sum-frequency generation in optics. 7. Continuous measurement of the relative phase of two Bose-Einstein condensates using light scattering We have demonstrated an experimental technique based on stimulated light scattering to continuously sample the relative phase of two spatially separated Bose-Einstein condensates of atoms [12]. This is the first time that the phase of a condensate could be determined in a non-destructive way. The phase measurement process created a relative phase between two condensates with no initial phase relation, read out the phase, and monitored the phase evolution. By monitoring the phase of two condensates at two separated times, interferometry between two trapped Bose-Einstein condensates without need for splitting or recombining. 8. High-Contrast Interference in a Thermal Cloud of Atoms Currently, there is considerable interest in characterizing the coherence properties of non-condensed systems including ultracold fermions, fermion pairs, and ultracold molecules. We have used an interferometric autocorrelation technique, previously applied only to condensates [13], to study the coherence properties of an alkali gas at finite temperature [14]. Bragg diffraction was used to create two spatially separated wave packets, which would interfere during expansion, analogous to a Young's double slit experiment. Fringe visibility greater than 90% was observed in a thermal cloud. We have shown that interference is lost when the separation between the wave packets exceeds the coherence length. However, the coherence length grows during ballistic expansion, and can become arbitrarily large. This can be understood by the conservation of local phase-space density, where the decrease in density is accompanied by a decrease in momentum spread. When the sample was filtered in momentum space using long, velocity-selective Bragg pulses, the contrast was enhanced; an effect the simple theory of a non-interacting gas could not account for. 1. F. Shimizu, Phys. Rev. Lett. 86, 987 (2001). 2. T.A. Pasquini, Y. Shin, C. Sanner, M. Saba, A. Schirotzek, D.E. Pritchard, and W. Ketterle, Phys. Rev. Lett. 93, 223201 (2004). 3. C.A. Stan, M.W. Zwierlein, C.H. Schunck, S.M.F. Raupach, and W. Ketterle, Phys. Rev. Lett. 93, 143001 (2004). 4. A.E. Leanhardt, A. Görlitz, A.P. Chikkatur, D. Kielpinski, Y. Shin, D.E. Pritchard, and W. Ketterle, Phys. Rev. Lett. 89, 190403 (2002). 5. Y. Shin, M. Saba, M. Vengalattore, T.A. Pasquini, C. Sanner, A.E. Leanhardt, M. Prentiss, D.E. Pritchard, and W. Ketterle, Phys. Rev. Lett. 93, 160406 (2004). 6. M. Möttönen, T. Mizushima, T. Isoshima, M.M. Salomaa, and K. Machida, Phys. Rev. A 68, 023611 (2003). 7. C.H. Schunck, M.W. Zwierlein, C.A. Stan, S.M.F. Raupach, W. Ketterle, A. Simoni, E. Tiesinga, C.J. Williams, and P.S. Julienne, Phys. Rev. A 71, 045601 (2005). 8. M.W. Zwierlein, C.A. Stan, C.H. Schunck, S.M.F. Raupach, A.J. Kerman, and W. Ketterle, Phys. Rev. Lett. 92, 120403 (2004). 9. C.A. Regal, M. Greiner, and D.S. Jin, Phys. Rev. Lett. 92, 040403 (2004). 10. J.R. Abo-Shaeer, D.E. Miller, J.K. Chin, K. Xu, T. Mukaiyama, and W. Ketterle, Phys. Rev. Lett. 94, 040405 (2005). 11. K. Xu, T. Mukaiyama, J.R. Abo-Shaeer, J.K. Chin, D.E. Miller, and W. Ketterle, Phys. Rev. Lett. 91, 210402 (2003). 12. M. Saba, T.A. Pasquini, C. Sanner, Y. Shin, W. Ketterle, and D.E. Pritchard, Science 307 (2005). 13. J.E. Simsarian, J. Denschlag, M. Edwards, C.W. Clark, L. Deng, E.W. Hagley, K. Helmerson, S.L. Rolston, and W.D. Phillips, Phys. Rev. Lett. 85, 2040–2043 (2000). 14. D.E. Miller, J.R. Anglin, J.R. Abo-Shaeer, K. Xu, J.K. Chin, and W. Ketterle, Phys. Rev. A 71, 043615 (2005).

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Articles in Peer-reviewed Journals	C. A. Stan and W. Ketterle "Multiple species atom source for laser-cooling experiments" Rev. Sci. Instrum. 76, 063113 (2005) , Jan-2005
Articles in Peer-reviewed Journals	D.E. Miller, J.R. Anglin, J.R. Abo-Shaeer, K. Xu, J.K. Chin and W. Ketterle "High-Contrast Interference in a Thermal Cloud of Atoms" Phys Rev. A 71, 043615 (2005) , Jan-2005
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